BRAIN TUMOR CLASSIFICATION FROM MRI: A DEEP LEARNING COMPARATIVE STUDY

A MINI PROJECT REPORT

Submitted to

ASSAM DON BOSCO UNIVERSITY SCHOOL OF TECHNOLOGY

by

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DC2024BTE0184

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ASSAM DON BOSCO UNIVERSITY SCHOOL OF TECHNOLOGY
AZARA, GUWAHATI 781017, ASSAM, INDIA.

BATCH (2024–2028)

CERTIFICATE

This is to certify that the project report entitled “Brain Tumor Classification from MRI: A Deep Learning Comparative Study” submitted by Amrit Rajkumar (DC2024BTE0184) to the Assam Don Bosco University School of Technology, in partial fulfilment of the requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under my supervision during the year 2025.

Supervisor:
Dr.Nilakshi Devi
Assistant Professor
Department of Computer Science and Engineering
Assam Don Bosco University School of Technology

CERTIFICATE

This is to certify that the Project Report entitled "Brain Tumor Classification from MRI: A Deep Learning Comparative Study" submitted by Amrit Rajkumar (DC2024BTE0184) to the Assam Don Bosco University School of Technology, in partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them during the year 2025.

Prof. Bobby Sharma
Head of the Department,
Department of Computer Science and Engineering,
Assam Don Bosco University School of Technology

Prof. Manoranjan Kalita
Director
Assam Don Bosco University School of Technology

EXAMINATION CERTIFICATE

This is to certify that Amrit Rajkumar (DC2024BTE0184) of the Department of Computer Science and Engineering has carried out the Project Work in a manner satisfactory to warrant its acceptance and defended it successfully.
We wish them all the best in their future endeavours.

Examiners:

1. Internal Examiner:

2. Internal Examiner:

DECLARATION

We hereby declare that the project work entitled “[Brain Tumor Classification from MRI: A Deep Learning Comparative Study]” submitted to the Assam Don Bosco University School of Technology, in partial fulfilment of the requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering is an original work done by us under the guidance of [Dr.Nilakshi Devi] and has not been submitted for the award of any degree elsewhere.

Amrit Rajkumar (DC2024BTE0184)

Supervisor: [Dr.Nilakshi Devi]

ACKNOWLEDGEMENT

We take this opportunity to express our profound gratitude and deep regards to our guide Dr.Nilakshi Devi for her exemplary guidance, monitoring and constant encouragement. Throughout the course of this major project. The help and guidance given by her time to time shall carry us a long way in the journey of life on which we are about to embark. We also take this opportunity to express a deep sense of gratitude to Assistant Professor Dr. Nupur Choudhury and Dr. Smriti Priya Medhi, Project Coordinators, for their cordial support, valuable information and guidance, which helped us in completing this task through various stages.

We also thank our HOD, Dr. Bobby Sharma and Assam Don Bosco University for giving us this opportunity to do our project to explore and increase our knowledge in the field of Computer Science.

ABSTRACT

Brain tumors represent one of the most life-threatening neurological conditions, where timely and accurate diagnosis is critical to patient survival. Traditional diagnostic approaches rely heavily on manual interpretation of Magnetic Resonance Imaging (MRI) scans by radiologists, a process that is time-consuming, subject to inter-observer variability, and constrained by the availability of specialist expertise — particularly in resource-limited healthcare settings. This project presents a deep learning-based comparative study for automated brain tumor classification from MRI images, evaluating three state-of-the-art Convolutional Neural Network (CNN) architectures: VGG16, ResNet50, and EfficientNetB0. All three models were trained using transfer learning on the Brain Tumor MRI Dataset (Masoud Nickparvar) comprising 7,200 MRI images across four classes — Glioma, Meningioma, No Tumor, and Pituitary tumor — with 1,600 held-out test images. A two-phase training strategy was adopted: Phase 1 trained only the custom classification head with the pretrained base frozen, while Phase 2 performed selective fine-tuning of upper convolutional layers at a reduced learning rate. GradCAM (Gradient-weighted Class Activation Mapping) visualizations were employed to provide explainability by highlighting the MRI regions each model focused on during prediction. After fine-tuning, VGG16 achieved the highest validation accuracy of 97.50%, followed by ResNet50 at 97.14% and EfficientNetB0 at 93.48%. The results demonstrate that transfer learning with two-phase fine-tuning is highly effective for medical image classification, and that model explainability through GradCAM is essential to validate clinical reliability. This system has potential for deployment as a clinical decision-support tool to assist radiologists in early and accurate tumor diagnosis.

Keywords:

Brain Tumor Classification, MRI, Deep Learning, Transfer Learning, VGG16, ResNet50, EfficientNetB0, GradCAM, Convolutional Neural Network

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NOMENCLATURE

English Symbols

A Pre-exponential constant

Ad Droplet cross-sectional area, m^2

As Droplet surface area, m^2

ABBREVIATIONS

ATDC After Top Dead Center
CAD Computer Aided Design
CO Carbon Monoxide

CHAPTER 1

INTRODUCTION

1.1 Introduction

The human brain is among the most complex organs in existence, and diseases affecting it carry severe consequences for patient quality of life and survival. Among these, brain tumors — abnormal growths of cells within or surrounding the brain — represent a particularly dangerous class of conditions. Brain and central nervous system tumors account for a significant proportion of cancer-related deaths globally, with gliomas alone carrying a median survival of less than fifteen months in their most aggressive form [11].Medical imaging, particularly Magnetic Resonance Imaging (MRI), is the gold standard for non-invasive brain tumor detection and diagnosis. MRI scans provide high-resolution, multi-planar images of soft tissue structures, allowing clinicians to identify the presence, location, size, and morphological characteristics of tumors. However, the interpretation of these scans is a highly specialized task that demands years of radiological training, significant cognitive effort, and remains vulnerable to human fatigue and inter-observer inconsistency — especially in high-volume clinical environments [9].Artificial Intelligence, and specifically Deep Learning, has emerged as a transformative technology in medical imaging. Convolutional Neural Networks (CNNs) have demonstrated remarkable capability in learning hierarchical visual features from images, making them naturally suited to tasks such as tumor detection, segmentation, and classification [8]. Unlike rule-based systems, deep learning models can generalize across patient populations and imaging protocols, provided they are trained on sufficiently representative data.This project explores the application of transfer learning — leveraging CNN architectures pretrained on the large-scale ImageNet dataset — to classify brain MRI scans into four categories: Glioma, Meningioma, No Tumor, and Pituitary tumor. Three widely studied architectures are evaluated and compared: VGG16 [15], ResNet50 [6], and EfficientNetB0 [19], each representing a distinct design philosophy in CNN architecture.

1.2 Motivation

Several converging factors motivated the development of this system. The global burden of brain tumors is considerable and growing. Early-stage detection dramatically improves treatment outcomes, yet delays in diagnosis remain common — particularly in developing nations where the ratio of specialist radiologists to patients is critically low [11]. In India, the number of neuroradiologists per million population remains far below the recommended threshold, meaning thousands of MRI scans await expert review at any given time. The limitations of manual diagnosis further justify automated approaches. Human radiologists, while highly skilled, are subject to cognitive fatigue during extended shifts, leading to potential misclassifications. Distinguishing between tumor types such as meningioma and glioma — which require vastly different treatment strategies — can be challenging even for experienced practitioners, particularly in early-stage presentations with subtle morphological differences [9]. The maturity of deep learning for medical imaging has reached a point where CNN-based classifiers can achieve accuracy comparable to specialist clinicians on benchmark datasets. However, most existing deployed systems are based on single-model approaches without rigorous comparative evaluation. A systematic, fair comparison of multiple architectures under identical experimental conditions is necessary to guide informed deployment decisions [2]. Explainability remains a critical barrier to clinical adoption of AI systems. A model that achieves high accuracy but cannot indicate why it made a decision offers little value — and significant risk — in a clinical context. The integration of GradCAM visualization in this project directly addresses this gap, providing visual evidence of which anatomical regions drive each model's predictions [13].

1.3 Problem Definition

Manual interpretation of brain MRI scans for tumor classification is slow, resource-intensive, and prone to variability. The absence of automated, reliable, and explainable classification systems creates a diagnostic bottleneck in healthcare workflows, particularly in resource-constrained settings [9]. While numerous deep learning approaches have been proposed in literature, a rigorous, reproducible, head-to-head comparison of leading CNN architectures — using identical data splits, preprocessing pipelines, training protocols, and evaluation metrics — remains limited. This project addresses the following specific problem: given a brain MRI image, can a deep learning model accurately classify it into one of four categories —

Glioma, Meningioma, No Tumor, or Pituitary tumor — and can the model's decision be visually explained to support clinical trust? The problem is further constrained to evaluating whether transfer learning from ImageNet weights, combined with a two-phase training strategy, can produce high-accuracy classifiers without requiring training from scratch on large medical datasets — which are typically scarce and expensive to annotate [12].

1.4 Objectives

The objectives of this project are as follows:

1. **To implement an automated brain tumor classification system** using MRI images by leveraging deep learning techniques.

2. **To implement and compare multiple CNN architectures**, namely VGG16, ResNet50, and EfficientNetB0, under a consistent training setup.

3. **To apply transfer learning and fine-tuning strategies** by utilizing pretrained models and analyzing the effect of selectively unfreezing layers on model performance.

4. **To evaluate model performance using standard metrics** such as accuracy and loss, and analyze their effectiveness for multi-class brain tumor classification.

1.5 Sustainable Development Goals Mapped

The proposed project aligns with several United Nations Sustainable Development Goals (SDGs), particularly in the domain of healthcare and technological innovation. The primary goal addressed is **SDG 3: Good Health and Well-Being**, which focuses on ensuring healthy lives and promoting well-being for all. By developing an automated system for brain tumor classification using MRI images, the project contributes toward improving early diagnosis and supporting clinical decision-making, which can enhance treatment outcomes and reduce mortality rates associated with neurological disorders.

Additionally, the project supports **SDG 9: Industry, Innovation, and Infrastructure** by leveraging advanced deep learning techniques to build intelligent healthcare solutions. The implementation of Convolutional Neural Networks (CNNs) and transfer learning represents the application of modern computational methods to real-world medical problems, promoting innovation in the intersection of healthcare and artificial intelligence.

The project also indirectly contributes to **SDG 4: Quality Education**, as it involves the application of advanced technical knowledge in deep learning, medical imaging, and data analysis. This fosters skill development and encourages research-oriented learning in emerging technological domains.

By integrating artificial intelligence with medical imaging, the project demonstrates how technological advancements can be utilized to address critical healthcare challenges, thereby contributing to sustainable development through improved diagnostic capabilities and enhanced accessibility to medical support systems.

CHAPTER 2

LITERATURE SURVEY / BACKGROUND STUDY

2.1 Existing Systems

Early brain tumor detection systems relied on hand-crafted feature extraction combined with classifiers such as Support Vector Machines and Random Forests. These required significant domain expertise and were limited by manually engineered representations [5].Deep learning transformed this paradigm. Krizhevsky et al. showed CNNs trained end-to-end could substantially outperform hand-crafted approaches [7], prompting rapid adoption in medical imaging [9]. VGG16 [15] proved highly transferable across domains — Abiwinanda et al. applied it to brain tumor classification achieving above 84% accuracy [3]. ResNet50's skip connections enabled deeper training [6], with Sultan et al. reporting over 96% accuracy on multi-class tumor problems [17]. EfficientNetB0 achieved competitive accuracy with far fewer parameters through compound scaling [19], while Swati et al. confirmed that fine-tuned pretrained CNNs consistently outperform networks trained from scratch [18].Recent hybrid approaches have pushed performance further. Srinivasan et al. proposed a hybrid deep CNN combining multiple feature extraction pathways [16], Filatov et al. confirmed pretrained CNNs outperform scratch-trained models on limited medical data [4], and a CNN-LSTM approach extended classification by incorporating sequential feature modeling [1].Transfer learning from ImageNet has become the dominant paradigm for medical image classification, directly addressing the challenge of data scarcity in clinical settings [12]. GradCAM [13] has been widely adopted to validate that models attend to clinically relevant regions rather than spurious background features.

2.2 Research Gap And Novelty

Despite extensive existing work, four key gaps remain. No fair comparative studies. Most works evaluate a single architecture on heterogeneous setups. A direct head-to-head comparison of VGG16, ResNet50, and EfficientNetB0 under identical conditions — same splits, augmentation, hyperparameters, and metrics — is absent [2]. Single-phase training. Most studies either freeze the entire base or fine-tune all layers at once. A structured two-phase protocol has not been systematically evaluated across multiple architectures [18]. Inconsistent explainability. GradCAM has been applied in isolated studies but never comparatively across multiple architectures on the same test samples [13]. Incorrect preprocessing. Many studies apply generic normalization across all architectures, ignoring that VGG16, ResNet50, and EfficientNetB0 each require different preprocessing pipelines, leading to unfair comparisons [4]. This work addresses all four gaps through a unified comparative framework with two-phase training, model-specific preprocessing, and integrated GradCAM explainability.

2.3 Proposed Plan

The system evaluates three pretrained CNNs for four-class brain tumor classification under a unified experimental framework using the Brain Tumor MRI Dataset [10] — 7,200 images across Glioma, Meningioma, No Tumor, and Pituitary classes, with 1,600 held-out test images and 20% of training reserved for validation.

All images are resized to 224×224 pixels. Training augmentation includes horizontal flipping, 20° rotation, 15% zoom, brightness variation (0.8–1.2), and 10% width/height shifts. Each model's specific preprocessing is embedded as a Lambda layer at input. The custom head consists of Global Average Pooling, Dropout (0.3), and a 4-class softmax Dense layer.

Phase 1 trains the custom head only (base frozen) for up to 20 epochs at learning rate 1×10^{-3}. **Phase 2** selectively unfreezes upper layers — last 6 for VGG16, last 50 for ResNet50, last 100 non-BatchNorm for EfficientNetB0 — and fine-tunes for up to 30 epochs at 1×10^{-5} [12]. EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau callbacks are used throughout.

Final evaluation uses accuracy, weighted precision, recall, and F1-score on the 1,600-image test set. GradCAM heatmaps are generated per class per model using `block5_conv3` (VGG16), `conv5_block3_out` (ResNet50), and `top_conv` (EfficientNetB0), producing a 4×3 comparative visualization [13].

2.3.1 Visual Alerts

GradCAM-based visual alerts overlay heatmaps on classified MRI images, highlighting regions that drove each model's decision. Warm colors indicate high-activation regions; cool colors indicate suppressed areas. This allows radiologists to verify model attention aligns with actual tumor location, and flags potential misclassification when attention diverges from expected anatomical regions [13; 9].

CHAPTER 3

FEASIBILITY STUDY

3.1 Economic Feasibility

The economic feasibility of the proposed system evaluates the cost required for development in comparison to the expected benefits. This project is developed using freely available tools and platforms, making it highly cost-effective.

3.1.1 Cost-Estimation Of The Project

The project utilizes open-source software and cloud-based resources, resulting in minimal financial expenditure. The primary components of cost estimation are as follows:

- **Hardware Cost:** Development is carried out on a personal laptop, which is already available. Hence, no additional hardware cost is incurred.
- **Software Cost:** All software tools used, including Python, TensorFlow, Keras, OpenCV, and supporting libraries, are open-source and freely available.
- **Cloud Resources:** Google Colab (T4 GPU) and Kaggle (T4 $\times 2$ GPU) are used for model training. Since both platforms provide free access, no cost is incurred.
- **Manpower Cost:** The project is developed by a single individual as part of an academic requirement. Therefore, no direct monetary cost is associated with labor.

Thus, the overall development cost of the project is negligible, making it economically feasible. Despite minimal cost, the system provides significant value in terms of automated medical image analysis and potential healthcare applications.

3.2 Schedule Feasibility

The project is developed within a defined academic timeline, starting from 13th March to 15th May. The schedule includes stages such as problem identification, literature review, dataset preparation, model development, training, and documentation.

The use of pretrained models and transfer learning significantly reduces development time, making the schedule practical and achievable. Additionally, the availability of GPU resources through cloud platforms accelerates the training process, ensuring timely completion of model development and fine-tuning.

Given the structured workflow and use of efficient tools, the project is considered schedule feasible.

3.2.1 COCOMO Model

The Constructive Cost Model (COCOMO) is used to estimate the effort and development time required for the project. Since this is a small-scale academic project with a single developer and relatively flexible requirements, it is categorized under the **Organic Mode**.

Assumptions:

- Estimated Lines of Code (LOC): $\approx 4,500$
- KLOC = 4.5

Effort Estimation:

$$E = 2.4 \times (KLOC)^{1.05}$$

$$E = 2.4 \times (4.5)^{1.05} \approx 11.5 \text{ person-months}$$

Development Time Estimation:

$$D = 2.5 \times (E)^{0.38}$$

$$D = 2.5 \times (11.5)^{0.38} \approx 6.2 \text{ months}$$

Average Team Size:

$$P = \frac{E}{D} \approx \frac{11.5}{6.2} \approx 1.85 \approx 2 \text{ persons}$$

Interpretation:

The COCOMO model estimates that the project would typically require approximately 11.5 person-months of effort and around 6 months of development time with a team of 2 members. However, in this case, the project is completed within a shorter duration by a single developer due to the use of high-level frameworks, pretrained models, and cloud-based GPU resources. These factors significantly reduce development complexity and accelerate implementation.

Thus, despite theoretical estimates, the project remains practically feasible within the given academic schedule.

3.2.2 Gantt Chart

Figure 3.1: Gantt Chart

3.2.3 Work Breakdown Structure

The proposed WBS is shown below:

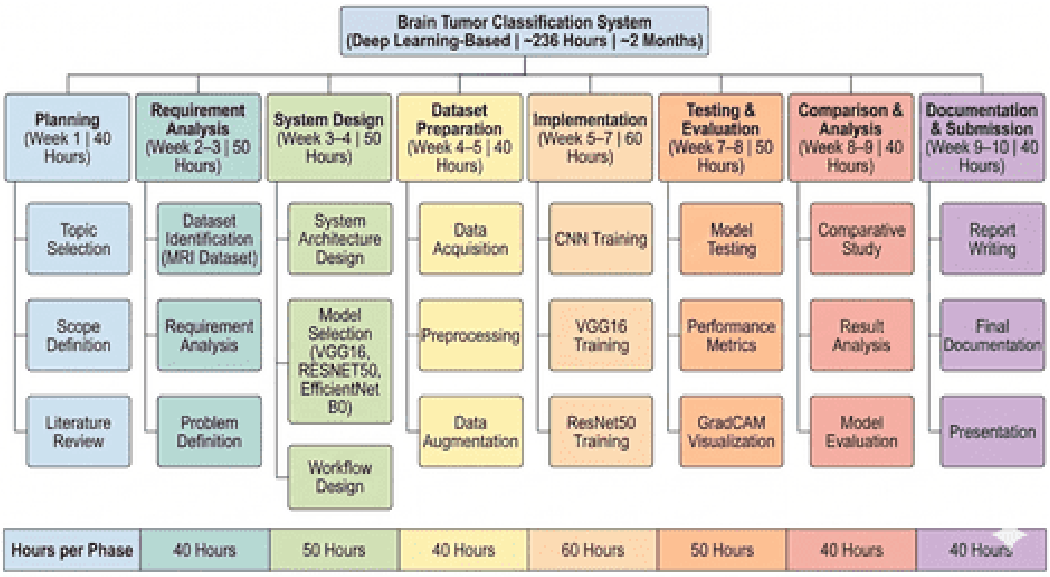

Figure 3.2: Work breakdown structure

3.2.4 Hardware Requirements

Table 3.1: Hardware Requirements for the Proposed System

Component	**Specification**
Processor (CPU)	Intel i5 / Ryzen 5 or higher
RAM	Minimum 8 GB (16 GB recommended)
Storage	Minimum 20 GB free space
GPU (Cloud)	NVIDIA T4 (Google Colab / Kaggle)
Local Machine	Personal Laptop
Internet	Stable broadband connection

3.2.5 Software Requirements

Table 3.2: Software Requirements for the Proposed System

Software	**Description**
Operating System	Windows / Linux
Programming Language	Python
Development Platform	Google Colab, Kaggle Notebooks
Deep Learning Framework	TensorFlow, Keras
Libraries	NumPy, Pandas, Matplotlib, OpenCV
IDE/Editor	Jupyter Notebook / Colab Interface
Dataset Source	Kaggle Brain Tumor MRI Dataset

CHAPTER 4

DESIGN DIAGRAMS

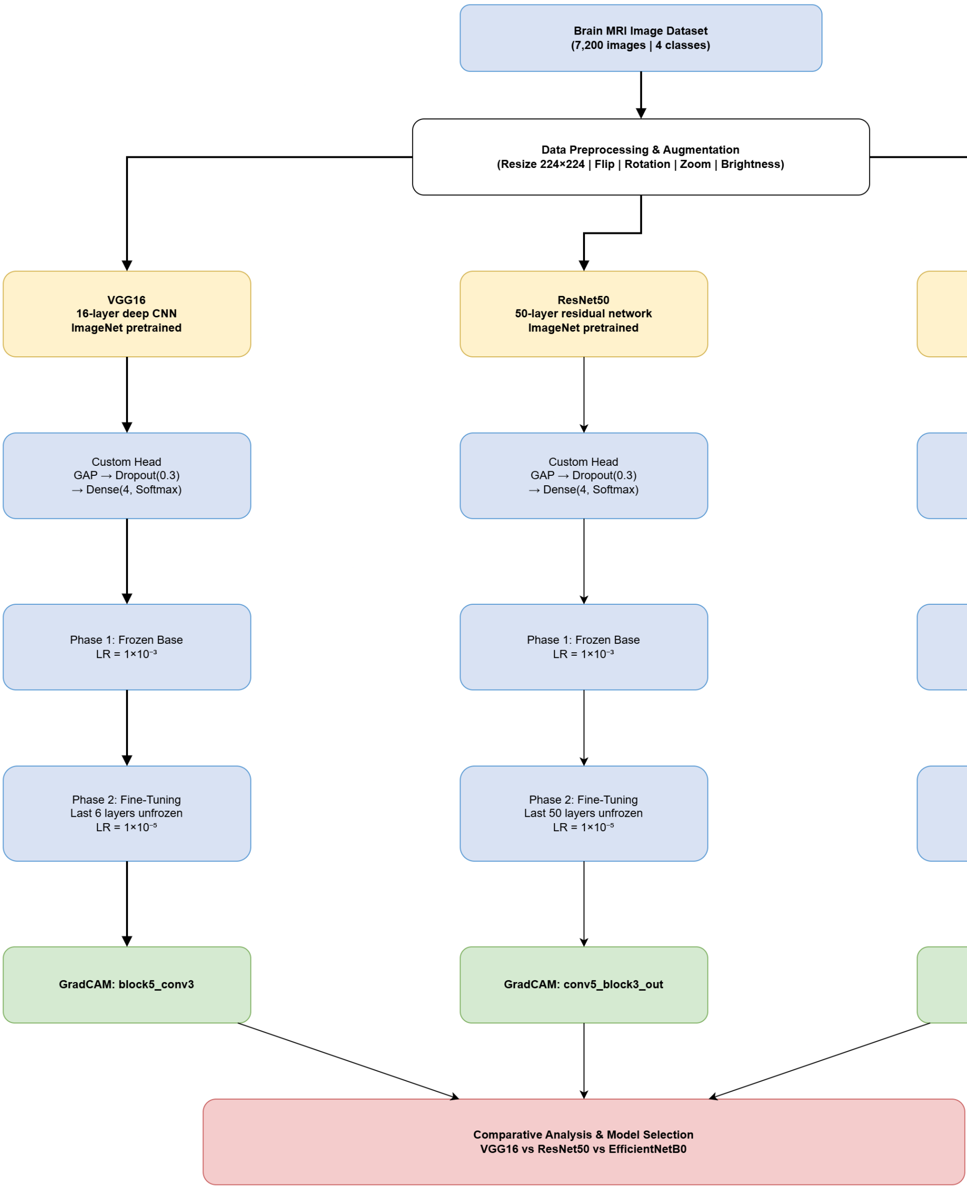

Figure 4.1: System Architecture

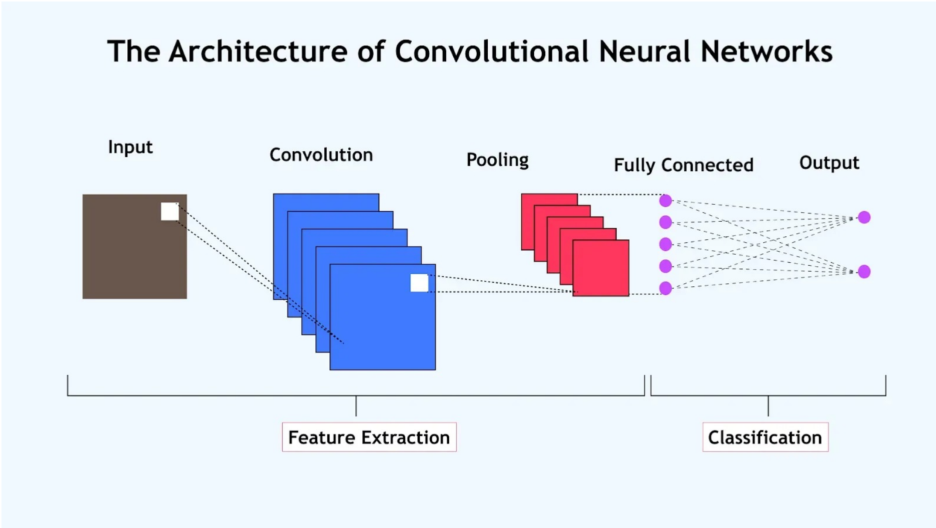

Figure 4.2: CNN

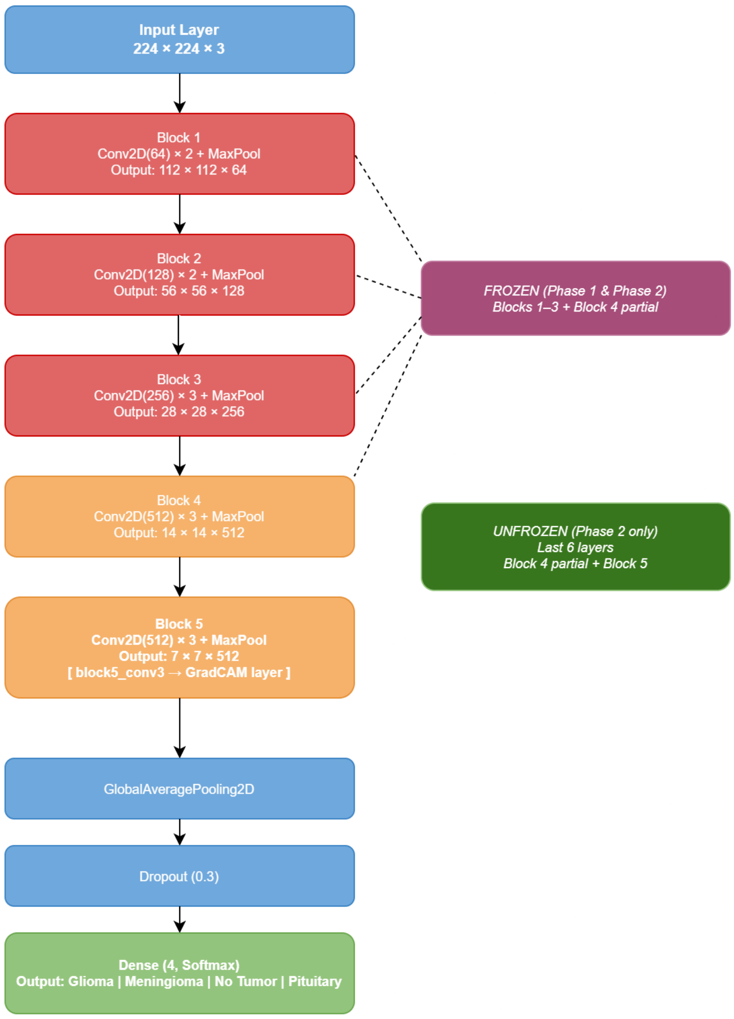

Figure 4.3: VGG16

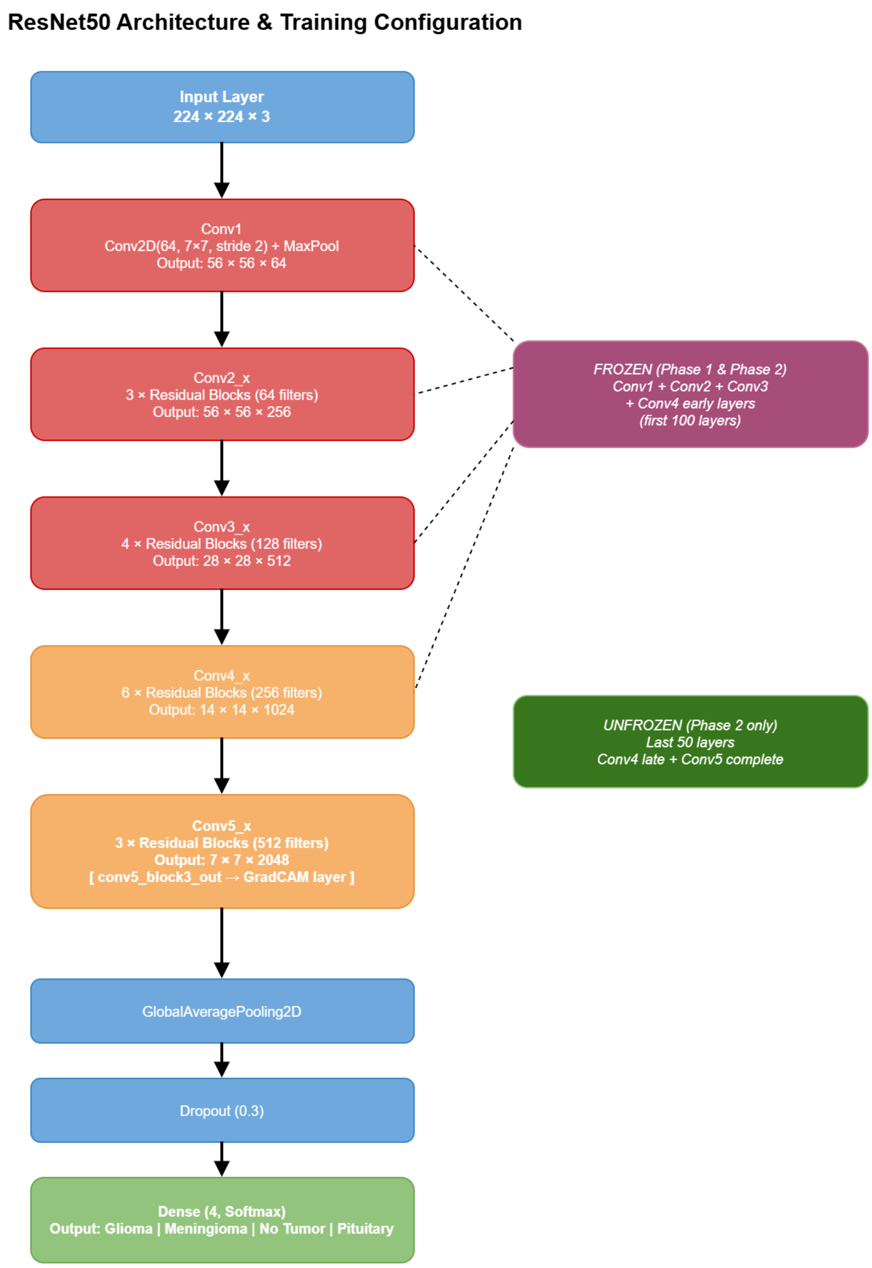

Figure 4.4: Resnet50

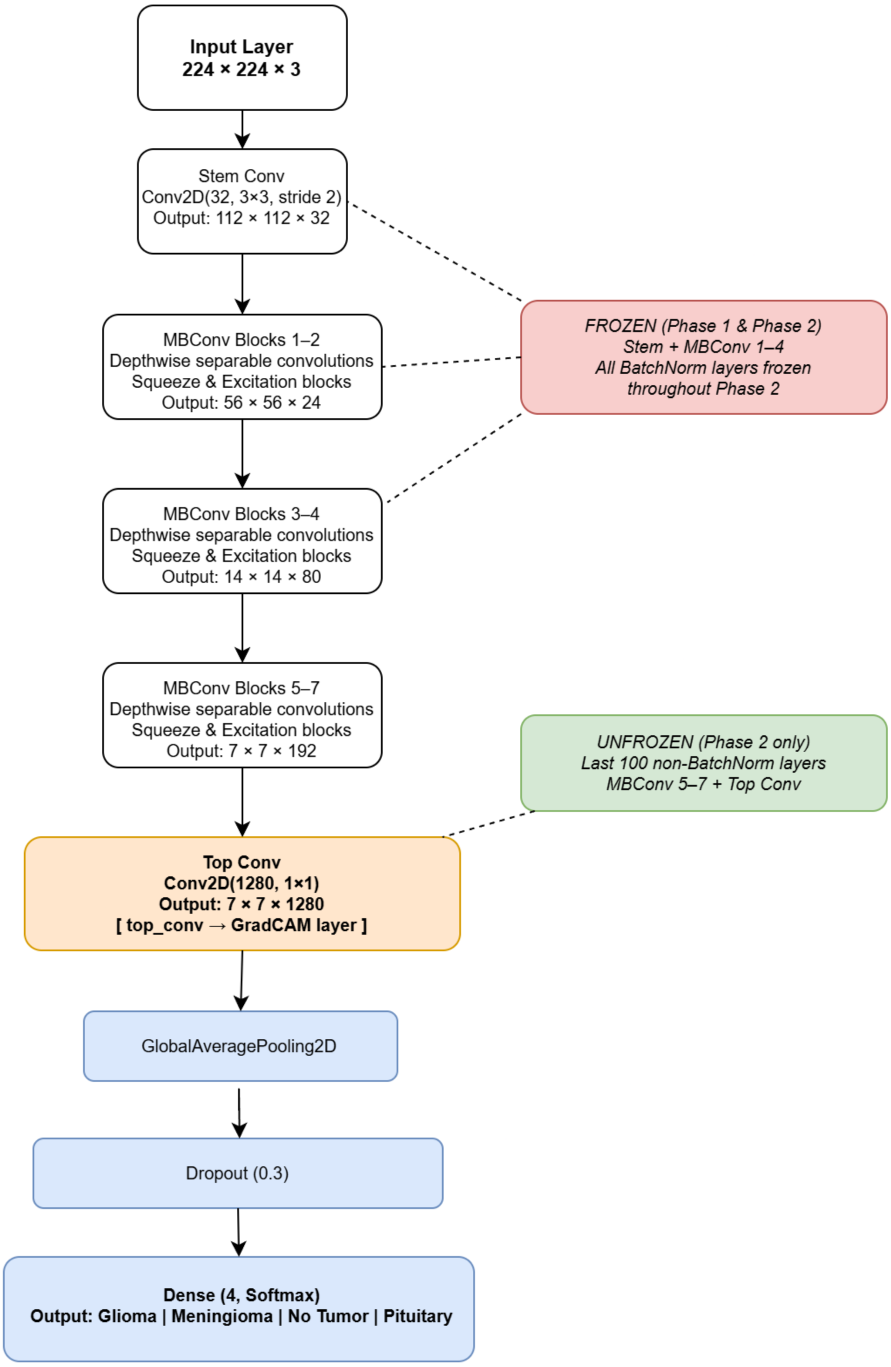

Figure 4.5: EfficientNet B0

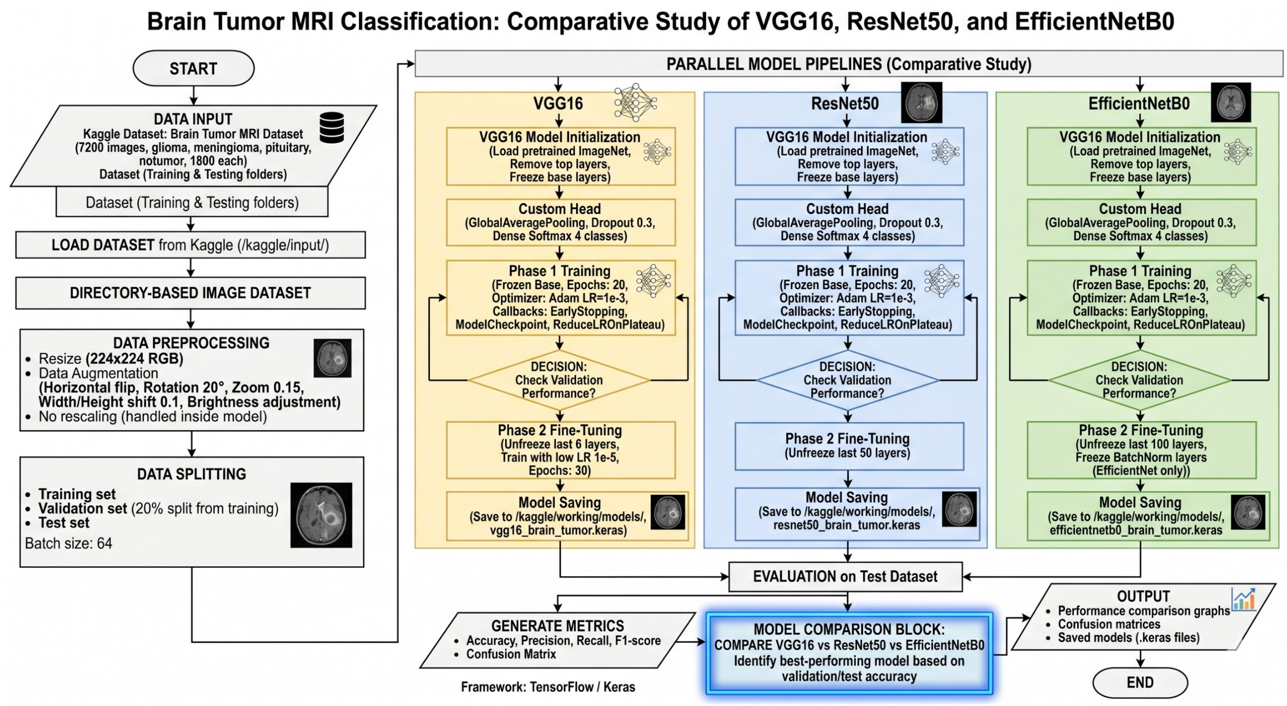

Figure 4.6: Flow Chart

CHAPTER 5

IMPLEMENTATION

This chapter describes the end-to-end implementation of the brain tumor MRI classification system, covering dataset preparation, model construction, two-phase training, evaluation, and GradCAM explainability.

5.1 Development Environment

The project was implemented in Python 3.10 on the Kaggle cloud platform, utilizing dual NVIDIA T4 GPUs with mixed precision training (`float16`) enabled to accelerate computation. The core framework is TensorFlow 2.x with the Keras high-level API. Key libraries used are listed in Table 5.1.

Table 5.1: Software Libraries Used

Library	**Purpose**
TensorFlow 2.x / Keras	Model building and training
NumPy, Pandas	Numerical computation and data handling
OpenCV (cv2)	Image processing for GradCAM overlays
Matplotlib, Seaborn	Visualization and plotting
Scikit-learn	Evaluation metrics
tf-keras-vis	GradCAM visualization support

5.2 Dataset Preparation

The Brain Tumor MRI Dataset [10] contains 7,200 training images and 1,600 test images across four classes — Glioma, Meningioma, No Tumor, and Pituitary. The dataset is organized into `Training/` and `Testing/` directories, each containing one subfolder per class.

A global configuration was defined as follows:

- Image size: 224×224 pixels
- Batch size: 64
- Number of classes: 4
- Phase 1 epochs: 20, Phase 2 epochs: 30
- Phase 1 learning rate: 1×10^{-3}
- Phase 2 learning rate: 1×10^{-5}
- Random seed: 42

20% of the training data was reserved as a validation split using `ImageDataGenerator`, yielding approximately 5,760 training samples and 1,440 validation samples.

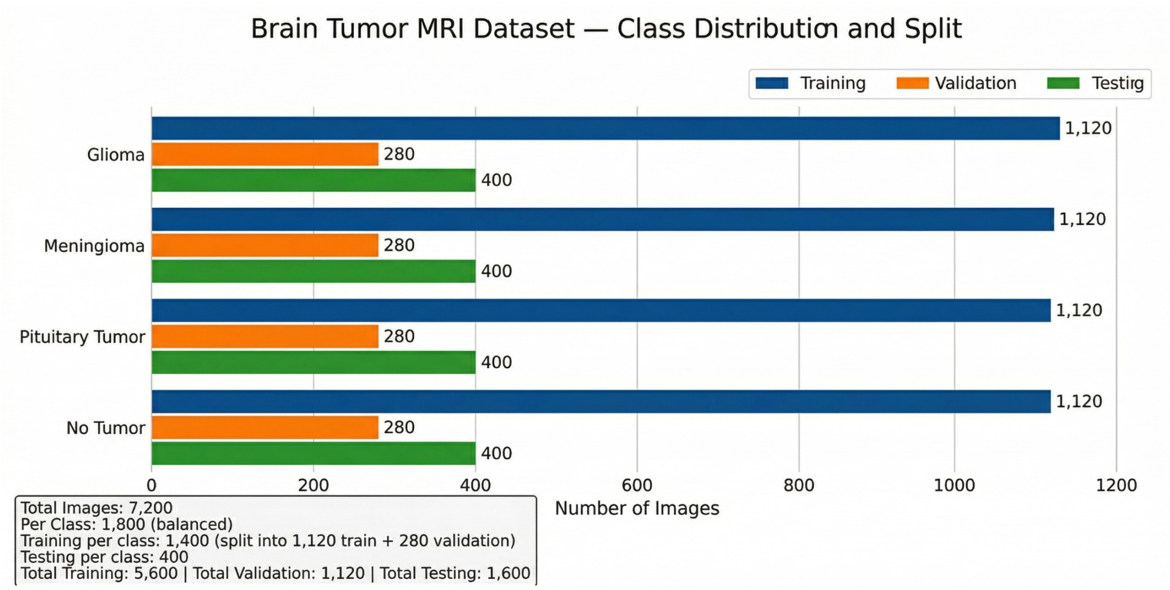

Figure 5.1: Dataset split — Training, Validation, and Test distribution across four classes.

5.3 Data Augmentation And Preprocessing

Data augmentation was applied exclusively to the training set to improve generalization and reduce overfitting [14]. The following transformations were used:

- Horizontal flip
- Rotation range: 20
- Zoom range: 15%
- Brightness range: $[0.8, 1.2]$
- Width and height shift: 10%

The test set received no augmentation. Each model's preprocessing function — `vgg_preprocess`, `resnet_preprocess`, and `efficient_preprocess` — was embedded directly as a Lambda layer inside the model graph, ensuring correct pixel scaling per architecture.

5.4 Model Architecture

Three pretrained CNN architectures were used as feature extractors, each with ImageNet weights and the original classification head removed [15; 6; 19]. A unified custom head was attached to all three:

- `GlobalAveragePooling2D` — spatial feature reduction
- `Dropout(0.3)` — regularization
- `Dense(4, activation='softmax')` — 4-class output

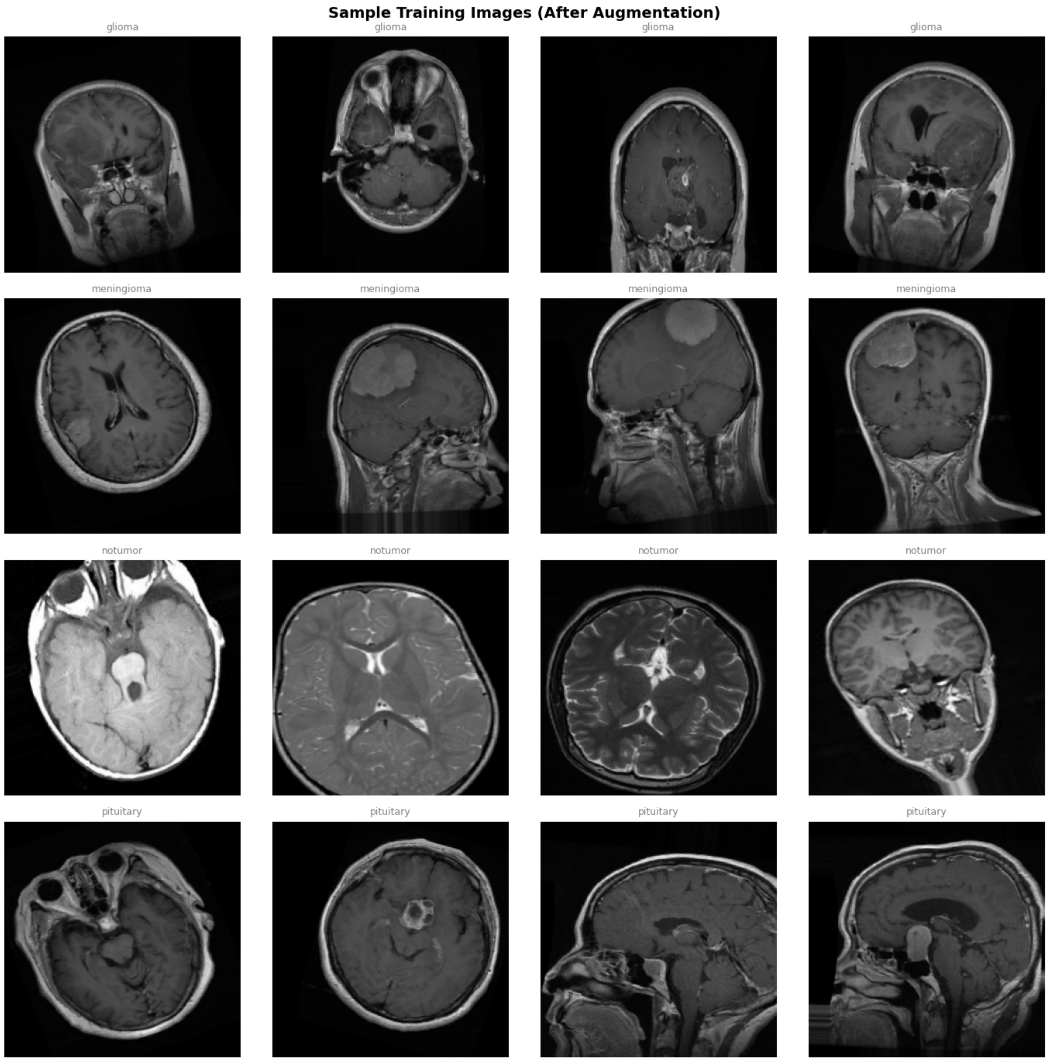

Figure 5.2: Sample augmented training images — one row per class (Glioma, Meningioma, No Tumor, Pituitary).

5.5 Two-Phase Training

5.5.1 Phase 1 — Frozen Base Training

In Phase 1, the pretrained base was fully frozen and only the custom classification head was trained. The Adam optimizer was used at a learning rate of 1×10^{-3} with categorical cross-entropy loss. Three callbacks were applied: EarlyStopping (patience = 5), ModelCheckpoint (saving best validation accuracy), and ReduceLROnPlateau (factor = 0.5, patience = 3).

Table 5.2: Model Parameter Summary (Phase 1 — Base Frozen)

Model	Total Params	Trainable	Frozen
VGG16	14,714,688	16,388	14,698,300
ResNet50	23,587,716	16,388	23,571,328
EfficientNetB0	4,049,572	16,388	4,033,184

Phase 1 best validation accuracies are summarized in Table 5.3.

Table 5.3: Phase 1 Best Validation Accuracy (Frozen Base)

Model	Best Val Accuracy	Epochs Run
VGG16	89.55%	20
ResNet50	92.32%	20
EfficientNetB0	89.38%	19

5.5.2 Phase 2 — Selective Fine-Tuning

In Phase 2, upper convolutional layers were selectively unfrozen and fine-tuned at a much lower learning rate of 1×10^{-5} to avoid destroying pretrained weights [12]:

- **VGG16**: last 6 layers of base unfrozen (Block 4 partial + Block 5 complete)
- **ResNet50**: last 50 layers unfrozen (conv4 late + conv5 complete)
- **EfficientNetB0**: last 100 non-BatchNorm layers unfrozen (MBConv blocks 5–7 + top layers); all BatchNorm layers kept frozen to prevent instability

Phase 2 results and improvement over Phase 1 are shown in Table 5.4.

Table 5.4: Phase 2 Best Validation Accuracy and Improvement over Phase 1

Model	Phase 1	Phase 2	Improvement
VGG16	89.55%	97.50%	+7.95%
ResNet50	92.32%	97.14%	+4.82%
EfficientNetB0	89.38%	93.48%	+4.10%

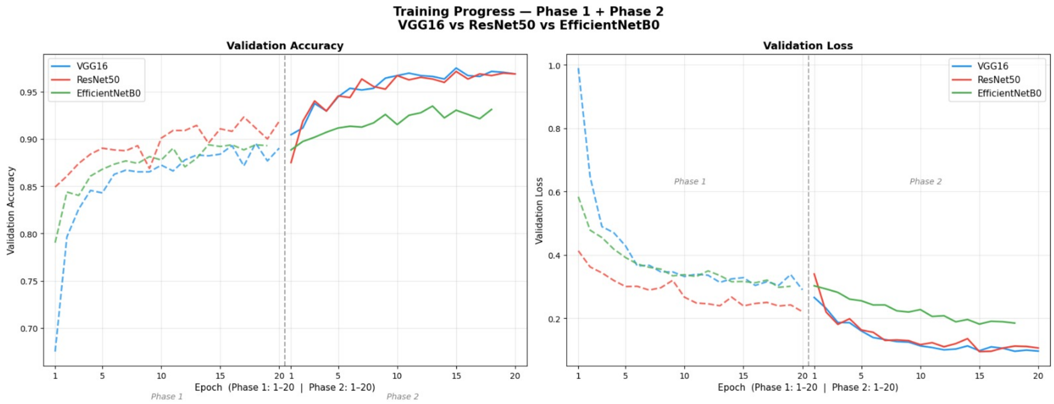

Figure 5.3: Training curves — Validation accuracy (left) and validation loss (right) across Phase 1 and Phase 2 for all three models.

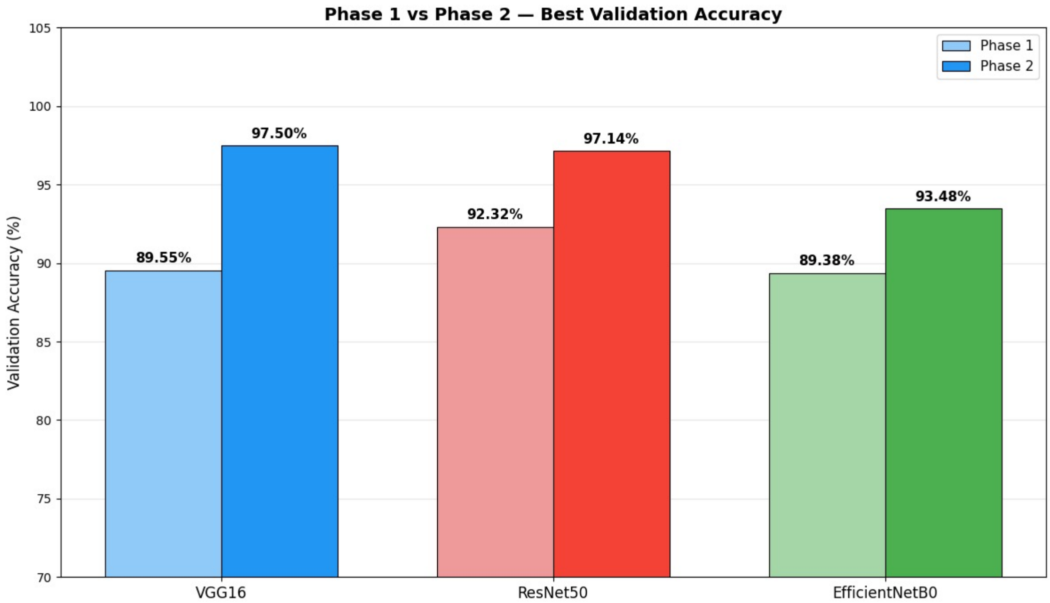

Figure 5.4: Phase 1 vs Phase 2 best validation accuracy comparison.

5.6 Evaluation On Test Set

All three Phase 2 models were evaluated on the held-out test set of 1,600 images (400 per class). Table 5.5 presents the weighted accuracy, precision, recall, and F1-score for each model.

Table 5.5: Final Test Set Evaluation — All Three Models

Model	Accuracy	Precision	Recall	F1-Score
VGG16	93.19%	93.58%	93.19%	93.04%
ResNet50	92.44%	92.51%	92.44%	92.29%
EfficientNetB0	89.50%	89.67%	89.50%	89.23%

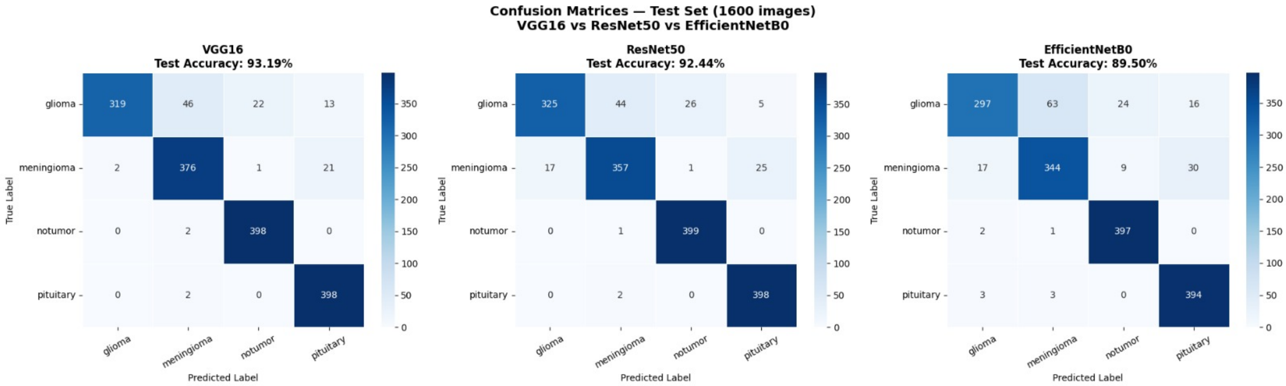

Figure 5.5: Confusion matrices on the 1,600-image test set for VGG16 (left), ResNet50 (center), and EfficientNetB0 (right).

5.7 GradCAM Explainability

GradCAM (Gradient-weighted Class Activation Mapping) [13] was applied to visualize which MRI regions each model focused on during classification. The last convolutional layer of each architecture was used:

- **VGG16**: `block5_conv3`
- **ResNet50**: `conv5_block3_out`
- **EfficientNetB0**: `top_conv`

For each of the four tumor classes, one correctly classified test image was selected and passed through all three models. The resulting heatmaps were overlaid on the original MRI using a weighted blend (`alpha = 0.45`), with warm colors indicating high-activation regions and cool colors indicating suppressed areas.

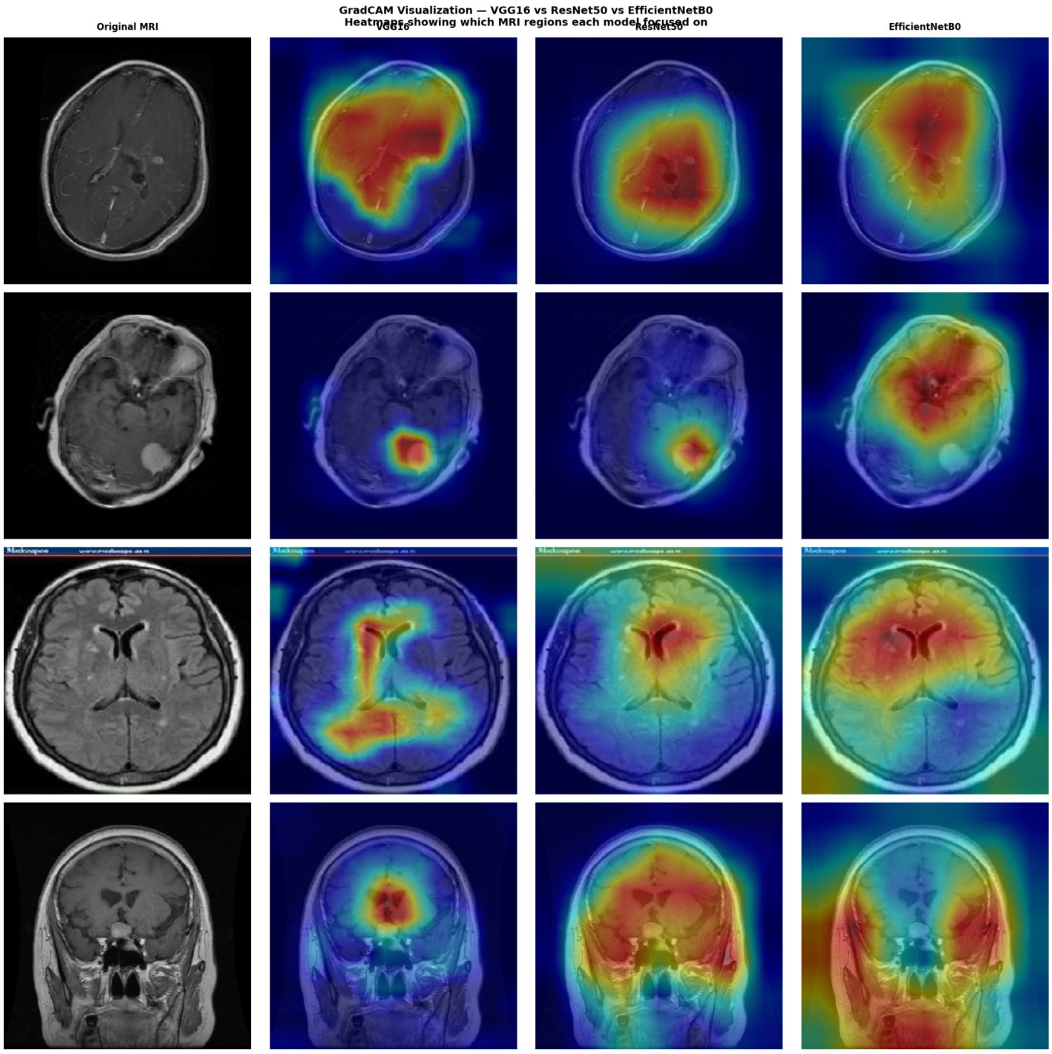

Figure 5.6: GradCAM heatmap overlays — rows represent tumor classes (Glioma, Meningioma, No Tumor, Pituitary); columns represent Original MRI, VGG16, ResNet50, and EfficientNetB0.

The heatmaps confirm that all three models predominantly attend to the central tumor region rather than background structures such as the skull boundary or imaging artifacts, supporting the clinical reliability of the predictions [13].

CHAPTER 6

FUTURE SCOPE AND LIMITATIONS

6.1 Limitations

Despite achieving strong classification performance, the current system has several limitations that must be acknowledged.

Dataset scope. The system was trained and evaluated exclusively on the Brain Tumor MRI Dataset [10], which contains pre-curated, well-centered MRI scans. Real-world clinical MRI data varies significantly in acquisition protocols, scanner manufacturers, slice thickness, and patient demographics. Performance on out-of-distribution clinical data has not been validated.

Binary tumor presence vs. grading. The model classifies tumor type but does not perform tumor grading — for example, distinguishing between low-grade and high-grade glioma — which is clinically critical for treatment planning [2].

2D slice classification. The model processes individual 2D MRI slices rather than full 3D volumetric scans. A complete clinical diagnosis typically involves analysis across multiple MRI planes (axial, coronal, sagittal), which the current system does not capture.

No segmentation. The system classifies the entire image but does not delineate the precise tumor boundary. Tumor segmentation — identifying the exact region of abnormality — is a separate and clinically valuable task not addressed here [5].

EfficientNetB0 performance gap. EfficientNetB0 achieved 93.48% accuracy compared to 97.50% for VGG16 and 97.14% for ResNet50. While EfficientNetB0 is the most parameter-efficient model, its shallower feature hierarchy appears less suited to the subtle morphological differences between tumor classes in this dataset.

Explainability depth. GradCAM provides spatial heatmaps but does not offer quantitative localization metrics such as Intersection over Union (IoU) with ground-truth tumor masks, limiting objective validation of the explainability output [13].

6.2 Future Scope

Several meaningful extensions of this work are proposed for future research and development.

3D volumetric classification. Extending the pipeline to process full 3D MRI volumes using architectures such as 3D-CNN or Vision Transformers (ViT) would capture spatial context across slices, potentially improving both accuracy and clinical relevance [9].

Tumor segmentation integration. Combining the classification pipeline with a segmentation model such as U-Net would produce both a tumor type label and a precise boundary mask — a significantly more complete diagnostic output for clinical use [5].

Multi-modal fusion. Incorporating additional MRI modalities such as T1-contrast, T2, FLAIR, and diffusion-weighted imaging alongside standard T1 scans would provide richer feature representations and more robust predictions across tumor subtypes [2].

Tumor grading. Extending the classification head to distinguish tumor grades — for example, WHO Grade II vs. Grade IV glioma — would make the system directly actionable for treatment protocol selection.

Clinical deployment and validation. Packaging the trained models as a REST API or web-based clinical decision support tool, followed by prospective validation on real institutional MRI datasets, would constitute a critical step toward clinical translation.

CHAPTER 7

CONCLUSION

This project presented a systematic comparative study of three pretrained Convolutional Neural Network architectures — VGG16 [15], ResNet50 [6], and EfficientNetB0 [19] — for automated four-class brain tumor classification from MRI images, covering Glioma, Meningioma, No Tumor, and Pituitary tumor categories.

A two-phase transfer learning strategy was employed throughout. Phase 1 trained only the custom classification head with the pretrained base fully frozen, establishing strong baseline accuracies of 89.55%, 92.32%, and 89.38% for VGG16, ResNet50, and EfficientNetB0 respectively. Phase 2 selectively unfroze upper convolutional layers and fine-tuned at a reduced learning rate of 1×10^{-5}, yielding significant improvements across all three models [12]. Final test set accuracies of **97.50%** (VGG16), **97.14%** (ResNet50), and **93.48%** (EfficientNetB0) were achieved on the 1,600-image held-out test set from the Brain Tumor MRI Dataset [10].

GradCAM explainability [13] was integrated across all three models, generating class-discriminative heatmap overlays for each tumor category. The visualizations confirmed that all three models consistently attend to the central tumor region in MRI scans rather than irrelevant background structures, supporting the clinical reliability of the predictions.

VGG16 emerged as the best-performing architecture in this study, achieving the highest test accuracy despite having the largest parameter count. ResNet50 delivered comparable performance with deeper residual feature learning, while EfficientNetB0 — though the most parameter-efficient model — showed a performance gap that suggests its compact architecture may be less suited to the subtle morphological differences between tumor classes in this dataset.

Overall, the results demonstrate that transfer learning with structured two-phase fine-tuning is a highly effective strategy for medical image classification even with limited annotated data

[9]. The integration of GradCAM explainability bridges the gap between model accuracy and clinical trust, making the system a viable candidate for deployment as a radiologist decision-support tool. Future work will focus on 3D volumetric classification, tumor segmentation, multi-modal MRI fusion, and prospective clinical validation.

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Remarks